

Moh Khalid Hasan, Ph.D.

Assistant Professor of Computer Science

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Experience

James Madison University, Assistant Professor, Department of Computer Science
Tenure-track faculty appointment in computer science.

Harrisonburg, VA, USA
Aug 2025 – present
1 year

- Conducts research in wireless communications, cybersecurity, trustworthy machine learning, and secure networked systems.
- Teaches undergraduate courses and supervises undergraduate research and capstone projects.

AISECLAB, Stevens Institute of Technology, Graduate Research Assistant
Research in wireless security and learning-driven communication systems.

Hoboken, NJ, USA
Feb 2021 – June 2025
4 years 5 months

- Investigated physical-layer security, privacy, cognitive radio, dynamic spectrum sharing, NOMA, UAV-assisted networks, and RIS-enabled communications.
- Developed analytical and learning-based methods for secure and efficient wireless resource allocation.

WiComAI Lab, Kookmin University, Research Associate
Research in optical wireless communication and intelligent networked systems.

Seoul, South Korea
Sept 2019 – Jan 2021
1 year 5 months

- Worked on optical camera communication, hybrid optical/RF networking, machine-learning-assisted communication systems, and IoT applications.

WiComAI Lab, Kookmin University, Graduate Research Assistant
Graduate research in optical camera and visible-light communication.

Seoul, South Korea
Sept 2017 – Aug 2019
2 years

- Developed and evaluated communication systems for localization, healthcare, transportation, and mobile robotic applications.

Education

Ph.D. Stevens Institute of Technology, Electrical Engineering

Hoboken, NJ, USA
Jan 2021 – Jan 2025

- Advisor: Min Song, Ph.D.; Co-advisor: Shucheng Yu, Ph.D.
- Dissertation: Toward Secure and Efficient Cognitive Radio Networks: Adaptive PHY Security and Learning-Driven Optimization

M.S. Kookmin University, Electronics Engineering

Seoul, South Korea
Jan 2017 – Jan 2019

B.S. Khulna University of Engineering & Technology, Electrical and Electronic Engineering

Khulna, Bangladesh
Jan 2013 – Jan 2017

Research Areas

Wireless Security and Privacy: Physical-layer security, privacy enhancement, secure NOMA, cognitive radio, and dynamic spectrum sharing.

Intelligent Wireless Networks: UAV-assisted and RIS-enabled communications, resource allocation, spectrum sharing, and emerging 6G systems.

Trustworthy Artificial Intelligence: Federated learning, privacy-preserving learning, adversarial machine learning, and machine learning for wireless optimization.

Optical Communications: Optical camera communication, visible-light communication, hybrid optical/RF systems, and intelligent transportation applications.

Publications

- Channel-Adaptive Privacy Enhancement for NOMA-Based Antagonistic Overlay Cognitive Networks** Feb 2026
Moh Khalid Hasan, Shucheng Yu, and Min Song. Vol. 75, no. 2, pp. 2910–2926.
Moh Khalid Hasan, Shucheng Yu, Min Song
[10.1109/TVT.2025.3606837](#) (IEEE Transactions on Vehicular Technology)
- U-Net-Based Optical Camera Communication in Intensity-Constrained Scenarios** May 2025
Md Osman Ali, **Moh Khalid Hasan**, Yitong Wang, Mohamed Shehata, Jing Fu, Stanley Luong, and Ke Wang. Vol. 43, no. 10, pp. 4712–4721.
Md Osman Ali, **Moh Khalid Hasan**, Yitong Wang, Mohamed Shehata, Jing Fu, Stanley Luong, Ke Wang
[10.1109/JLT.2025.3543281](#) (Journal of Lightwave Technology)
- Uplink Secrecy in RIS-Aided MIMO-NOMA Networks with User-Centric Artificial Noise** Jan 2025
Moh Khalid Hasan, Shucheng Yu, and Min Song. IEEE ICC, pp. 2617–2622.
Moh Khalid Hasan, Shucheng Yu, Min Song
(IEEE International Conference on Communications)
- Cooperative NOMA-Based Spectrum Leasing With Multiple Secondary Users** Jan 2023
Moh Khalid Hasan, Xiaochan Xue, Shucheng Yu, and Min Song. Vol. 72, no. 11, pp. 14543–14558.
Moh Khalid Hasan, Xiaochan Xue, Shucheng Yu, Min Song
[10.1109/TVT.2023.3287230](#) (IEEE Transactions on Vehicular Technology)
- Over-the-Air Federated Learning with Enhanced Privacy** Jan 2023
Xiaochan Xue, **Moh Khalid Hasan**, Shucheng Yu, Laxima Niure Kandel, and Min Song. IEEE ICC, pp. 4546–4551.
Xiaochan Xue, **Moh Khalid Hasan**, Shucheng Yu, Laxima Niure Kandel, Min Song
[10.1109/ICC45041.2023.10278765](#) (IEEE International Conference on Communications)
- Optical Camera Communication in Vehicular Applications — A Review** Jan 2022
Moh Khalid Hasan, Md Osman Ali, Md Habibur Rahman, Mostafa Zaman Chowdhury, and Yeong Min Jang. Vol. 23, no. 7, pp. 6260–6281.
Moh Khalid Hasan, Md Osman Ali, Md Habibur Rahman, Mostafa Zaman Chowdhury, Yeong Min Jang
[10.1109/TITS.2021.3086409](#) (IEEE Transactions on Intelligent Transportation Systems)
- Optical Wireless Hybrid Networks — Trends, Opportunities, Challenges, and Research Directions** Jan 2020
Mostafa Zaman Chowdhury, **Moh Khalid Hasan**, Md Shahjalal, Md Tanvir Hossan, and Yeong Min Jang. Vol. 22, no. 2, pp. 930–966.
Mostafa Zaman Chowdhury, **Moh Khalid Hasan**, Md Shahjalal, Md Tanvir Hossan, Yeong Min Jang
[10.1109/COMST.2020.2966855](#) (IEEE Communications Surveys & Tutorials)
- Simultaneous Data Transmission Using Multilevel LED in Hybrid OCC/LiFi System — Concept and Demonstration** Jan 2019
Moh Khalid Hasan, Nam Tuan Le, Md Shahjalal, Mostafa Zaman Chowdhury, and Yeong Min Jang. Vol. 23, no. 12, pp. 2296–2300.
Moh Khalid Hasan, Nam Tuan Le, Md Shahjalal, Mostafa Zaman Chowdhury, Yeong Min Jang
(IEEE Communications Letters)

Grants and Funding

- **Faculty Development Grant, James Madison University** — Principal Investigator; funded, 2025.
- **4-VA Collaborative Research Grant, James Madison University** — Principal Investigator; submitted, 2026.
- **CCI Cybersecurity and Critical Infrastructure Research Call, James Madison University** — Principal Investigator and Co-Principal Investigator on two submitted proposals, 2025.

Teaching Experience

- **IT 201: Computational Structures and Logic**, James Madison University — Undergraduate lecture and laboratory; Fall 2025 and Spring 2026.
- **IT 212: Digital Electronics**, James Madison University — Undergraduate lecture and laboratory; Spring 2026.
- **EE/CPE/AAI 595: Applied Machine Learning**, Stevens Institute of Technology — Graduate instructor; Fall 2024 and Spring 2025.
- **ENGR 232 / ENGR 212: Analog Circuits Laboratories**, Stevens Institute of Technology — Graduate teaching assistant and lead laboratory instructor; January 2022–May 2024.
- **Basic Electronic Circuits I Laboratory**, Kookmin University — Graduate teaching assistant; September–December 2019.

Student Mentoring and Supervision

- **Undergraduate Research Assistants, James Madison University, Spring 2026:** Matthew Engel and Korbinian Huber.
- **Undergraduate Capstone Students, James Madison University, Summer 2026:** Michael V. Castellano, Noah Reed, Brian Stock, William Jedrzejczak, Dilpreet Gill, and Cole Philip Walther.
- **M.S. Student Research Group, Stevens Institute of Technology, Spring 2025–Present:** Sudhanshu Kakkar, Fiza Pathan, and Azizul Haque.
- **Doctoral Student Peer Mentoring Program, Stevens Institute of Technology, 2022–2025:** Official peer mentor.
- **ECE Scholarship Program:** Academic mentor for master’s and undergraduate students.

Professional Service

- **Journal reviewer:** IEEE Transactions on Communications; IEEE Transactions on Vehicular Technology; IEEE Journal on Selected Areas in Communications; IEEE Communications Magazine; IEEE Access; Computer Networks; Scientific Reports; and other journals.
- **Technical program committee:** IEEE INFOCOM 2026 workshops EIN 2026 and ISAC-FutureG 2026.
- **Conference reviewer:** IEEE SM 2026, IEEE GCAIoT 2025, ICCIT 2025, and EICT 2025.
- **Conference service:** Lead Student Volunteer, IEEE INFOCOM 2023; Student Volunteer, ICUFN 2018 and KICS 2018.

Leadership

- **President, ECE Ph.D. Research Club, Stevens Institute of Technology** — March 2023–May 2025.
- **President, Bangladeshi Student Association, Stevens Institute of Technology** — June 2024–May 2025.

Honors and Awards

Third Prize, ECE Three Minute Thesis Competition Recognized for concise and effective research communication.	Stevens Institute of Technology Jan 2024
First Prize, Ph.D. Research Expo Recognized for excellence in doctoral research presentation.	Stevens Institute of Technology Jan 2023
Provost Doctoral Fellowship	Stevens Institute of Technology Jan 2021
Graduate School Academic Excellence Award	Kookmin University Jan 2019

Professional Development

- **Teaching at the College Level Certificate Program**, Stevens Institute of Technology.